Construction of statistical eyes

Fixed-rate folding and differential-clock triggering

The same received waveform produces substantially different eyes when its time reference changes. This report reconstructs both statistical eyes from the supplied ADS waveform record and verifies the bin counts independently. The triggered eye has a visibly larger opening because it follows the received clock rather than a constant nominal period.

Objective and setup

The assignment specifies a nominal 40 Gbps bit rate, so one unit interval (UI) is 25 ps and the displayed window is 2 UI = 50 ps. It describes a frequency deviation of approximately +/-2 MHz about 40 GHz with a 1 us modulation period. Such spread spectrum clocking (SSC) causes a fixed time reference to drift relative to data. [1]

Waveform record	Histogram grid	Common plot axes
1,639,001 measured samples	120 voltage x 50 time bins	Time: 0 to 50 ps
1 ps nominal spacing	dV = 10 mV; dt = 1 ps	Voltage: -0.1 to 1.1 V
Vo, CLKp and CLKn	Column normalization	Shared linear probability scale

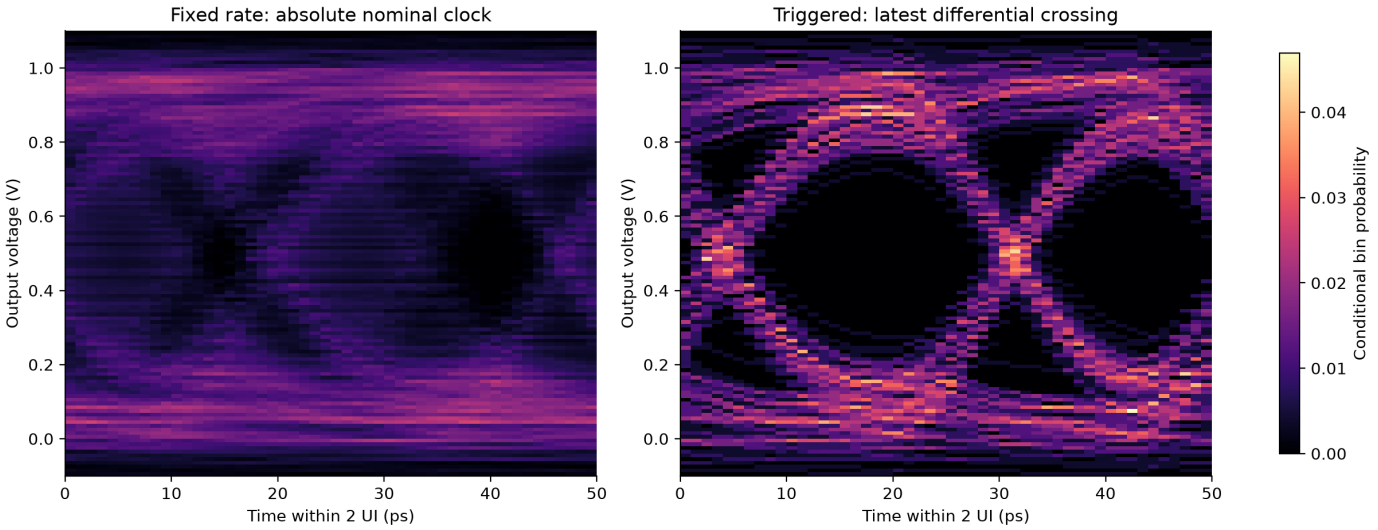


Figure 1. Regenerated fixed-rate and triggered eyes. Color denotes the probability of a voltage bin conditional on a time bin. Both panels use the same axes and color scale; differences are due to time assignment rather than voltage rescaling.

Main finding

The received clock crossing phase spans 15.007 ps relative to successive nominal 50 ps intervals. Fixed-rate folding spreads transitions across that changing phase. Resetting time at positive differential-clock crossings reduces this misalignment and reveals the opening seen in Figure 1. This comparison supports the assignment's SSC alignment argument; it does not establish a bit error rate.

Method and bin conventions

1. Preserve the waveform values

Read the four CSV columns Time, Vo, CLKp and CLKn. Require finite values and strictly increasing time, then express time in picoseconds. All voltage samples are retained unchanged. Interpolation is applied only to clock crossing times. The observed Vo range is -0.09617345 to 1.08581 V, inside the selected -0.1 to 1.1 V voltage range.

2. Fixed-rate folding

For each sample time t, use a fixed origin t0 = 0 ps and fold the sample into a 50 ps window. The corresponding integer bin is obtained by flooring the time in 1 ps units and then taking integer modulo 50:

```
j_fixed = floor((t - t0) / dt) mod 50
k = floor((Vo - vmin) / dV)
```

The voltage offset vmin is essential: a negative voltage belongs near the bottom of the histogram rather than wrapping around to a high-voltage row. Time and voltage bins are left-closed and right-open; the final voltage bin also accepts exact vmax. Only floating point roundoff near integer bin edges is snapped before flooring.

3. Differential-clock triggers

Form c = CLKp - CLKn. Detect a positive crossing between adjacent samples with c[n-1] < 0 and c[n] >= 0. Linear interpolation estimates its time:

```
a = -c[n-1] / (c[n] - c[n-1])
t_cross = t[n-1] + a * (t[n] - t[n-1])
```

For each data sample, choose the latest crossing at or before its time. Set age = t - t_cross and retain the sample only when 0 <= age < 50 ps. Then j_triggered = floor(age / dt). Triggered ages never wrap, and a sample is counted at most once. This avoids double counting where adjacent 2 UI windows overlap. Samples before the first crossing and in a gap past 50 ps are excluded.

4. Count and normalize vertically

Accumulate a count H[k,j] at each voltage/time bin. Normalize each occupied time column separately; an empty column remains zero without division by zero:

```
P[k,j] = H[k,j] / sum_k H[k,j]
sum_k P[k,j] = 1 (for occupied columns)
```

P is a discrete conditional bin probability P(voltage bin | time bin). A continuous voltage density would additionally divide by dV. The plotted values therefore depend on the chosen voltage and time bins. The vertical-normalization procedure follows the assignment starter and the eye-diagram lecture. [2,3]

Timing evidence and verification

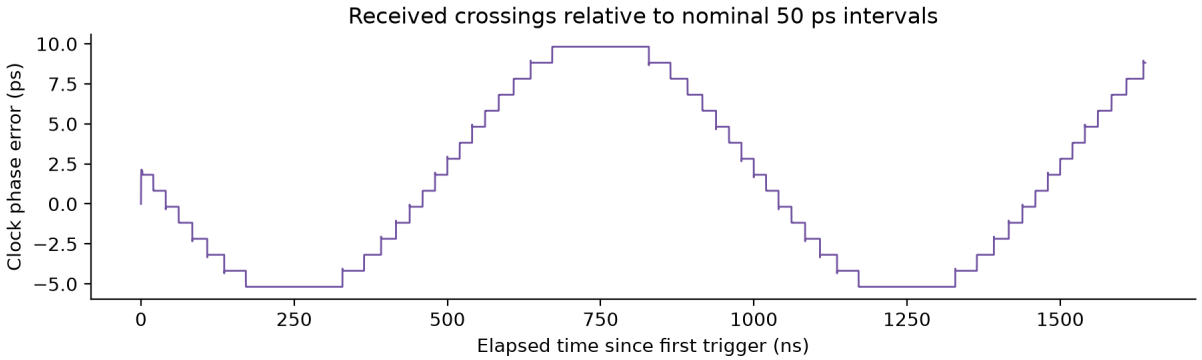


Figure 2. Positive crossing times relative to the first crossing plus n x 50 ps. This is received-clock phase error relative to a nominal reference; it includes the effects present in the supplied sampled waveform.

Checked result	Fixed rate	Triggered
Samples counted	1,639,001	1,638,788
Count per occupied time column	32,780 to 32,781	32,754 to 32,777
Occupied / total time columns	50 / 50	50 / 50
Largest column-sum error	2.22e-16	1.11e-16

There are 32,777 interpolated positive crossings. Their intervals range from 48.842 to 51.135 ps (mean 50.000269 ps). Triggering excludes 170 samples before the first crossing and 43 samples with age at least 50 ps, accounting for the difference in totals. Unequal column counts are therefore expected.

Verification

Fourteen unit tests passed, covering bin boundaries, negative voltage offsets, crossing interpolation, trigger gaps and overlap, normalization, invalid input, and synthetic clock-rate drift. A separately implemented full-record histogram check matched every integer bin: 6,000 of 6,000 fixed-rate bins and 6,000 of 6,000 triggered bins. All probabilities are finite, and occupied-column sums differ from one by at most 2.22e-16.

Reproduction

From the HW05 folder, install the analysis requirements and run the analysis, tests and independent check. The report builder reads the regenerated summary and figures and needs the separate report requirements.

```
python src/hw5.py
python -m unittest discover -s tests -v
python report/build_report.py
python tests/verify_delivery.py
```

Limits and conclusion

The triggered eye shows improved timing alignment for this finite ADS record. The observed crossing intervals are received-waveform measurements, not an estimate of the SSC source's frequency extrema. No noise model, BER extrapolation, threshold optimization or standardized eye-mask test was performed. Any eye opening inferred from these bins remains sample- and bin-dependent.

References: [1] HW5_stat_eye_construction.pdf, pp. 1-3. [2] HW5_stateye_construction_sample.py. [3] 5 Eye diagram.pdf, pp. 4-5 and 14-17. Data: HW5_ADS_waveforms.csv.